



HACK AWAY HUNGER 2026 · DSMHACK x CORTEVA AGRISCIENCE ·
SUBMISSION

Count Me In

Every hour counts. So do you.

Count Me In turns volunteering into verified hours, so Iowans facing SNAP's new 80-hour work rule keep their food assistance — and food pantries gain the regular volunteers they need.

“We don't create food. We stop it from disappearing.”

Live app	https://count-me-in-iowa.netlify.app
Live demo stage	https://count-me-in-iowa.netlify.app/stage
Verify an attestation	https://count-me-in-iowa.netlify.app/verify (no key needed)
Source code (MIT)	https://github.com/chessoren/hackawayhunger
Team	Oren Vidal
Prizes	Grand Prize · Community Impact Award · People's Choice Award

Submission at a glance

Volunteering → verified hours → SNAP kept. An SMS & voice assistant in 8 languages helps lowans meet the new 80-hour rule, and gives pantries the regular volunteers they need.

The problem

Since the July 2025 federal law (P.L. 119-21), adults 18–64 without a child under 14 must prove 80 hours a month of work, training or volunteering to keep SNAP; exemptions for veterans, people experiencing homelessness and former foster youth were removed. Iowa SNAP recipients fell 9% in a year (271,880 → 247,907, May 2025 → May 2026) while 1 in 8 lowans face food insecurity. The law already counts volunteering hour for hour — **what's missing is the proof.**

What we built

Person	SMS / voice / web assistant in 8 languages: letter photo → 3-sentence summary + deadline; 6-question screening decided by a cited, deterministic rules engine; 80-hour gauge; 3 accessible missions (walk, DART bus, schedule); QR or 4-digit check-in; consent-gated sending to the State; DELETE by SMS.
Host site	10-minute onboarding, printable QR poster, today's list (first name only), one-tap validation signed with Ed25519, no-show alert, grant-ready monthly report, audit log.
State	Monthly PDF attestation with QR; a caseworker verifies authenticity in one second; any modification breaks the signature.
Community	Anonymized impact dashboard ($k \geq 10$) and a map of opportunity deserts for Iowa's 99 counties; editable impact model: $\approx \$303,000$ of food aid preserved per year for 500 people supported.
Navigators	Human relay: HELP tickets, possible exemptions, people at risk of missing their month.

How to test in 2 minutes

1. Open **count-me-in-iowa.netlify.app/stage** and paste a Claude (sk-ant-...) or OpenAI (sk-...) API key — it stays in your browser.
2. On the phone, text **Hi** (or **Hola, Jambo**), then tap **"Send the sample State letter"**.
3. Answer the questions in your own words; pick mission 1, 2 or 3.
4. Use the demo clock: **Shift starts** → **Scan the QR at the door** → **Shift ends** → scan again → ✓ **Confirm** as the coordinator.
5. Text **ATTESTATION**, open the signed attestation, then visit **/dashboard** for the Iowa map.

About API keys (temporary). For the submission, Count Me In runs on each user's own AI key (Claude or OpenAI). The key never leaves the browser except to the chosen provider, and no key is shipped in the code. **If Count Me In wins, the prize will fund the AI, SMS and voice keys so the service runs for everyone** — people like Linda will never see a key; they just text a number.

About the demo video. Because no keys were shipped, the 2-minute walkthrough was recorded with pre-written AI responses standing in for the model's structured outputs; the live app runs on the real Claude or OpenAI APIs with your key. Site names, people and the sample letter are illustrative; no partnership is implied.

Built with

TypeScript · React · Vite · Node.js · Claude (Anthropic SDK) · OpenAI SDK · Netlify / Vercel functions · Twilio SMS & Voice webhooks · Web Speech API · Leaflet + GeoJSON · pdf-lib · qrcode · jsQR · TweetNaCl (Ed25519) · YAML rules · Zod · Vitest (20 tests) · Playwright · Fish Audio (video voice-over)



Count Me In

**The people in the food line can be the
ones who move it —
and keep their food aid doing it.**

Volunteering → verified hours → SNAP kept. By SMS or voice, in 8 languages.

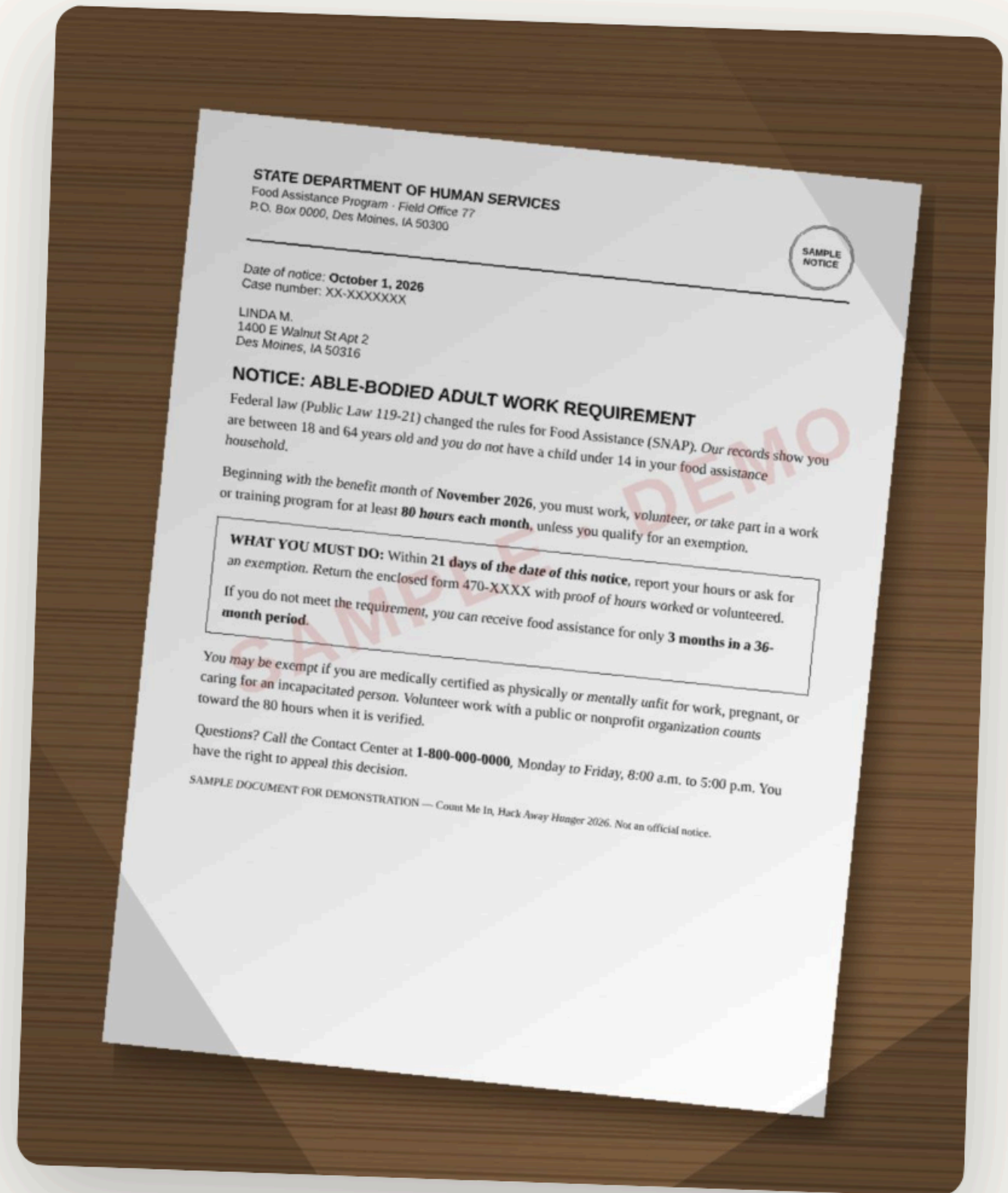
MEET LINDA

58. Des Moines. Cleans houses 10 h a week. No car.

In October a letter arrives: from now on, **80 hours a month** of work, training or volunteering — or her food assistance stops.

21 days to respond. A letter she can barely parse.

Without help, by spring she is in the line at a DMARC pantry that already breaks records.



WHAT CHANGED IN 2025

A federal rule just reached hundreds of thousands of Iowans

18 → 64

The 80-hour rule now covers adults up to 64 (was 54).

Child under 14

Parents of teenagers are no longer exempt.

Exemptions removed

Veterans, people experiencing homelessness, former foster youth.

3 months in 36

Without 80 h/month or an exemption, SNAP stops.

—**9%**

Iowa SNAP recipients, May 2025 → May 2026 (271,880 → 247,907)

1 in 8

Iowans food insecure · 73M-meal gap (Map the Meal Gap 2025)

THE REAL PROBLEM

It's not willingness. It's proof.

1. **They don't know** if they're covered — or already exempt.
2. **They can't read the letter** — long, English, tight deadline.
3. **They can't reach hours** — no car, irregular shifts.
4. **They can't prove** what they do — volunteering leaves no trace.
5. **They miss the deadline** — one missed report cuts the aid.

The other half

Pantries chronically lack **regular weekday volunteers** — and track attendance on paper they can't certify.

SNAP provides ~9 meals for every 1 from the charitable network. **Every person who loses SNAP lands in the pantry line.**

THE INSIGHT

The law already counts volunteering — hour for hour.

We built the missing proof.



The person

Keeps SNAP. No app, no password — in their language.



The pantry

Gains regular volunteers.
Drops the paper sign-in sheet.



The State

Gets clean, signed proof instead of incomplete files.

One hour of volunteering serves three people at once.



Count Me In

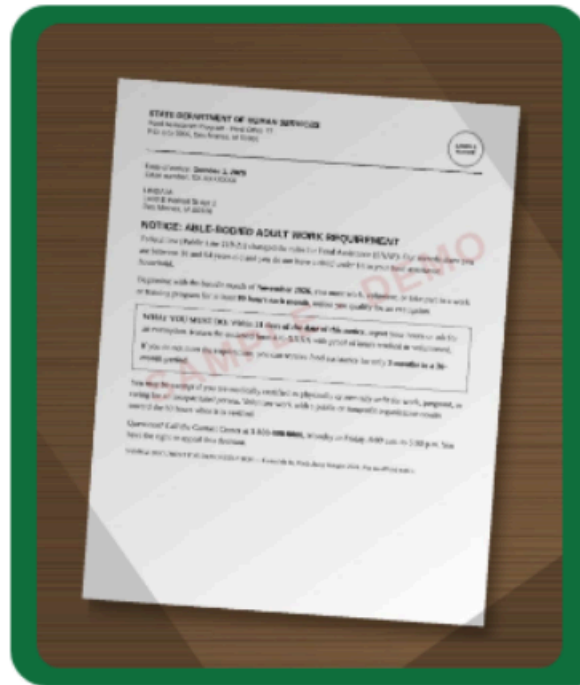
(515) 800-8080 · SMS · English



anytime to erase everything, or HELP to talk to a person.

I speak: English · Español · Kiswahili ·
العربية · العربية · Tiếng Việt · Français ·
Ikinyarwanda.

What's your first name?



Text message



Send the sample State letter (demo)

1 · THE LETTER, DECODED

Snap a photo. Get 3 plain sentences.

Text "Hi" to a number — SMS, voice call, or web. Photograph the State letter, even crooked.

Back, in her language: **what it says, the deadline, the one thing to do**, the number to call. A reminder is set 3 days before.

Vision + OCR with Claude or OpenAI · deadline computed from "within 21 days of the notice".

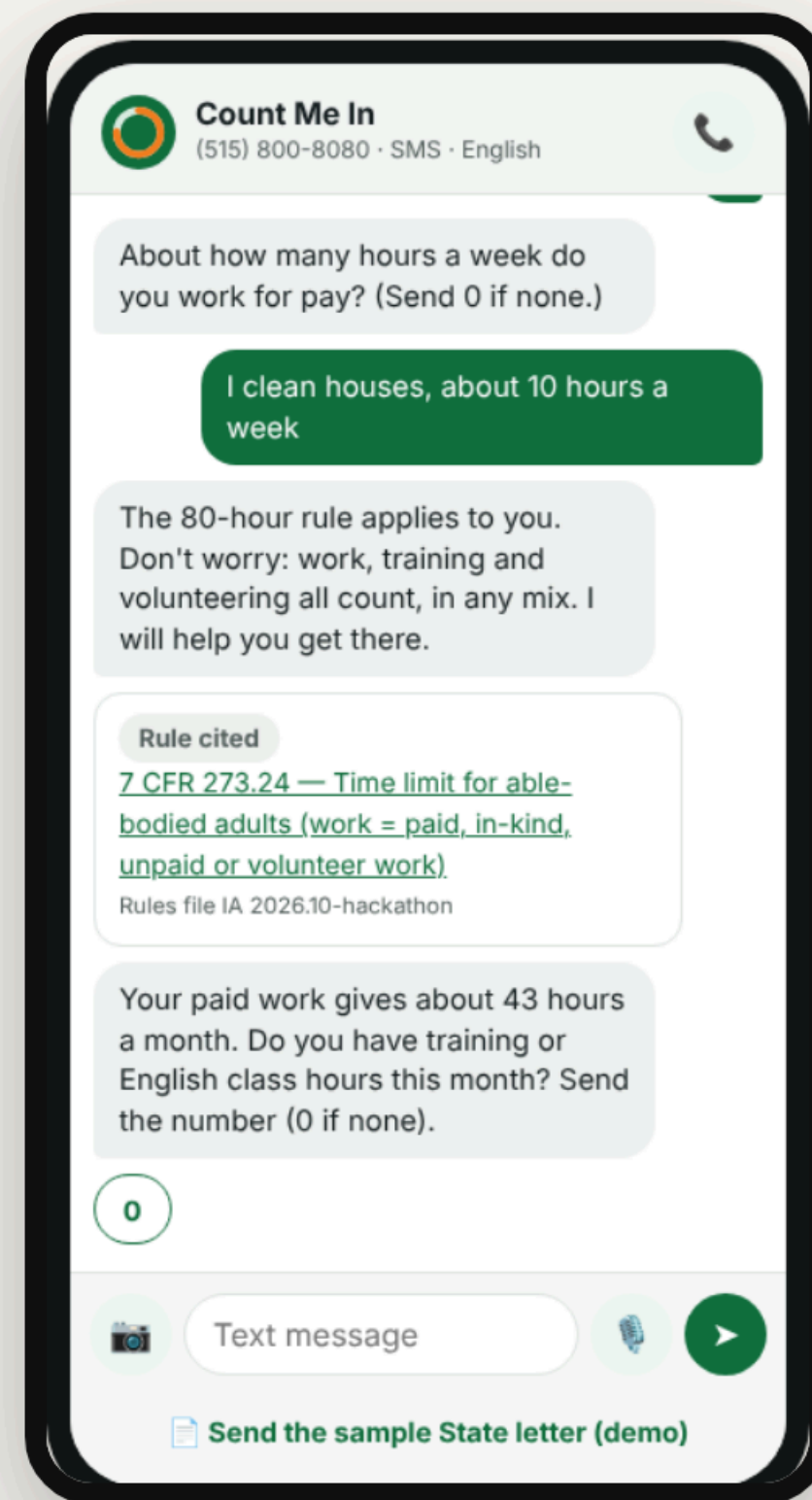
2 · "DOES THIS APPLY TO ME?"

6 questions. Her own words. A cited rule.

"My knees hurt a bit but I can still work." The AI turns words into facts — **a deterministic rules engine decides**, and cites the regulation.

Probably exempt → what proof to send. Covered → a plan.
Unsure → a human navigator.

Veteran? It explains, gently, that the exemption was removed in 2025.



I clean houses Monday and
Wednesday mornings

Here are missions that fit you:

1 Food distribution — Grace Church
Community Grocery

Tue, Oct 13, 09:00–13:00 (4 h) · 12
min walk

2 Food distribution — Eastside
Neighborhood Pantry

Sat, Oct 10, 08:00–12:00 (4 h) · 19
min walk

3 Welcome desk — Downtown
Community Kitchen

Sun, Oct 11, 12:00–15:00 (3 h) ·
Route 4 — E 14th St, about 33 min ·
seated work
Reply 1, 2 or 3.

1

2

3



Text message



[Send the sample State letter \(demo\)](#)

Count Me In · open source

3 · THREE MISSIONS, NEVER MORE

Ranked by real access, not volume

- **12-min walk** or a **DART bus line**
- Around her cleaning schedule
- Languages spoken on site, seated work
- Optimized to reach **80 h before the deadline**

She replies **"1"**. Booked. Reminder the evening before. At the door: **scan the QR** — or text the site's **4-digit code** from any basic phone.

The paper sign-in sheet, replaced

COORDINATOR • Pastor Dave

Grace Church Community Grocery

Today

Monthly report

QR poster

Audit log

Tuesday, October 13 — who's coming

Linda

Food distribution • 09:00–13:00

✓ 4 h confirmed

You only see a first name and a slot — never why someone volunteers, never their SNAP file. Validation is signed with this site's key (B8B9-698C-908B-1886).

10 minutes

to become a host
site + QR poster

One tap

to confirm hours,
signed Ed25519

First name only

never the SNAP file

Grant-ready

monthly report,
hours, value

An attestation that verifies itself in one second

✓ Authentic — not modified since it was signed

Name

Linda

Month

2026-10

Total hours

47 / 80

DATE	ORGANIZATION	MISSION	IN-OUT	HOURS	VALIDATED BY
2026-10-13	Grace Church Community Grocery	Food distribution	09:02–12:59	4	Pastor Dave key B8B9-698C-908B-1886
2026-10-01	Declared by the person	Paid work (self-declared, not verified by Count Me In)	—	43	—

Volunteering 4 h · Training 0 h · Paid work declared by the person 43 h. ID CMI-202610-EBBF56 · issued 10/13/2026, 12:59:02 PM · rules IA 2026.10-hackathon · platform key E726-FAB0-9CC3-7F5E

Download the PDF

Verify another document

✗ Not valid — Signature does not match the document. It was modified after signing, or not signed by Count Me In.

Name

Linda

Month

2026-10

Total hours

51 / 80

Change one digit → the signature breaks.

Double validation · hash-chained append-only ledger · coordinator + platform signatures · audit log.

Nothing goes to Iowa HHS without her explicit YES — logged, every time.

A map of opportunity deserts

County by county: car-free accessible volunteer hours ÷ hours needed by people under the rule.

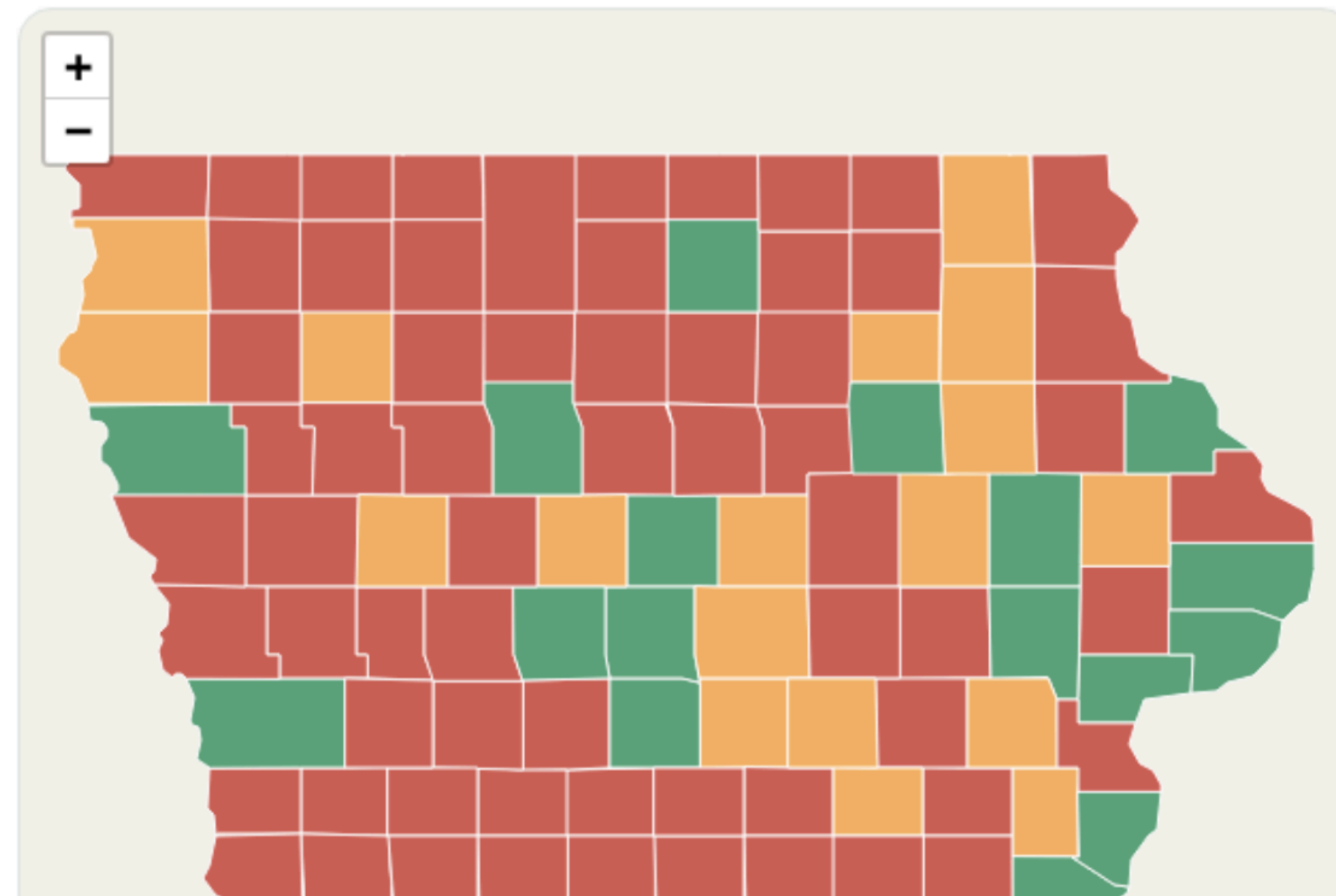
64 / 99

counties where 80 h are out of reach without a car (model)

Neutral data: where to open the next host site — or where to ask for a waiver. Never a cell under 10 people.

Where 80 hours are impossible without a car

For each of Iowa's 99 counties: car-free accessible mission hours ÷ hours needed by people subject to the rule. Red = the obligation is mathematically out of reach for part of the population.



IMPACT YOU CAN REDO IN YOUR HEAD

\$303,000

of food assistance preserved per year, for **500 people** supported

$500 \times 40\% \text{ would lose SNAP} \times 75\% \text{ kept with the tool} = \mathbf{150 \text{ people}}$ · $150 \times \$168.51 \text{ (Iowa avg, May 2026)} \times 12$

89,000

meals preserved (\$3.40/meal,
MMG 2025)

36,000

volunteer hours a year for host
sites

~\$5k

yearly running cost: SMS, AI,
hosting

HOW IT'S BUILT

AI understands. Rules decide. A human settles doubts.

The person

SMS · voice · web

The coordinator

QR · one-tap validation

Public & decision-makers

dashboard · map

Multilingual AI agent

Claude / OpenAI, structured outputs

Letter reader

vision → summary + deadline

Rules engine

one YAML per state, cited

Matching

walk · DART · schedule

Signed hours ledger

hash chain · Ed25519 · audit

Human relay

navigator console

Verifiable attestation

PDF + QR

Iowa HHS

only after a YES

Anonymized data

k ≥ 10 by county

React · TypeScript · Vite · Anthropic & OpenAI SDKs · Twilio SMS/Voice webhooks · Web Speech API · TweetNaCl · pdf-lib · Leaflet · Netlify/Vercel functions · Vitest · Playwright

DIGNITY · ACCESS · PRIVACY

Any phone. Any language. Always a human.

- **8 languages** + voice mode for people who read little
- Keywords & 4-digit codes work **without AI or data**
- 3 choices max · one question at a time · HELP always
- Minimal data · pseudonymous ledger · **DELETE by SMS**
- Volunteering is **one option**, never a condition for food



WHAT'S NEXT

MVP

Oct 2026 · full SMS
journey, QR,
attestation, map

DMARC pilot

Nov 2026 – Mar
2027 · 3–5 host sites
· EN + ES

All of Iowa

2027 · 55 counties ·
8 languages · Iowa
HHS link

Other states

2028 · one rules file
per state · open
source

Our ask: a partner pantry, a conversation with Iowa HHS on the attestation format, and a first corporate host site.

Hackathon build runs on the judge's own AI key (temporary). If we win, the prize funds the keys so the service runs for everyone.

We don't create food.
We stop it from disappearing.



ABOUT THE PROJECT

Inspiration

Linda is 58, lives in Des Moines, cleans houses ten hours a week and has no car. In October a letter tells her that, under the 2025 federal law, she must now show **80 hours a month** of work, training or volunteering, or lose her SNAP food assistance. She has 21 days.

Linda is a composite, but her situation is real for many Iowans. The law (P.L. 119-21) raised the SNAP work-rule age limit from 54 to 64, limited the parent exemption to households with a child under 14, and removed the exemptions for veterans, people experiencing homelessness and former foster youth. Iowa SNAP recipients fell **9% in a year** (271,880 → 247,907, May 2025 → May 2026), while **1 in 8 Iowans** face food insecurity. Everyone who loses SNAP ends up in a pantry line that is already breaking records.

Then we found the detail that shaped the project: **federal rules already count volunteering toward the 80 hours, hour for hour (7 CFR 273.24), as long as the hours are verified.** People don't lose SNAP because they refuse to work. They lose it because they can't prove what they do. Meanwhile pantries are short of regular weekday volunteers and track attendance on paper. So one well-recorded hour serves three people at once: **the person keeps their food aid, the pantry gains a volunteer, and the State gets clean proof.**

What it does

Count Me In turns volunteering into verified hours.

For the person — any phone, any of 8 languages. No app, no account: they text a number (or call, or use the web).

- **Letter decoded:** they photograph the State letter and get back three plain sentences, the deadline and the one thing to do, with an automatic reminder.
- **Screening:** six questions, one at a time, answered in their own words. A deterministic rules engine — not the AI — decides "probably exempt", "covered" or "let's check with a navigator", and cites the regulation.
- **Planning:** an 80-hour gauge, then at most **three missions** ranked by real access (walking time, DART bus, work schedule, language, physical limits).
- **Logging:** check-in by **QR code** or a **4-digit code** texted from any basic phone.
- **Always in control:** HELP reaches a human, DELETE erases everything.

For host sites. Pantries, churches, gardens and corporate volunteer programs onboard in about 10 minutes and print a QR poster. The coordinator sees only a first name and a time slot, never

the SNAP file. They confirm hours in one tap, and each confirmation is signed with the site's Ed25519 key. They get a monthly report ready for grant applications.

For the State. Each month the person gets an attestation as a PDF with a QR code. A caseworker scans it and knows in one second that it is authentic; **change a single digit and the signature breaks**. Nothing is sent to Iowa HHS without the person's explicit YES, and that consent is logged.

For the community. A public, anonymized dashboard (no cell under 10 people) and a **map of opportunity deserts**: for each of Iowa's 99 counties, how many volunteer hours are reachable without a car compared with the hours needed. Under our model, 64 counties, mostly rural, fall short.

Impact you can check in your head: 500 people supported $\times$ 40% who would lose SNAP $\times$ 75% who keep it with the tool = 150 people. $150 \times \$168.51$ (Iowa average benefit, May 2026) $\times$ 12 $\approx$ **\$303,000 of food aid preserved per year**, plus about 36,000 volunteer hours for host sites. All of this runs on a few thousand dollars a year.

How we built it

- **React + TypeScript + Vite**, mobile-first and accessible: large touch targets, high contrast, right-to-left Arabic, a Burmese font, and a voice mode using the Web Speech API.
- **One channel-agnostic agent** — the same code answers the web phone and the **Twilio** SMS and Voice webhooks. Keywords and site codes are handled first, without AI, so basic phones and outages still work. AI is used only for free text.
- **AI layer:** Claude (Anthropic SDK) or OpenAI, always with **structured JSON outputs**. It does three things: understand free text, read letter photos with vision and OCR, and translate.
- **Rules engine:** `rules/iowa.yaml`, one readable file per state with its sources. When the law changes, you edit the file, not the code.
- **Signed ledger:** append-only and hash-chained, with Ed25519 signatures (TweetNaCl). It stores a salted pseudonym, never a phone number.
- **Attestations:** signed server-side, built with **pdf-lib**, a QR code, and a verification page that also works by camera scan (**jsQR**).
- **Map:** **Leaflet** with Iowa county GeoJSON.
- **Hosting:** serverless functions on **Netlify**.
- **Quality:** **Vitest** (20 tests, including forgery and tampering cases) and a full **Playwright** end-to-end run.

API keys — temporary. For the judging, the app asks each user for their own Claude or OpenAI key. The key stays in the browser and is never in the code. **If we win, the prize will fund the AI, SMS and voice keys so the service runs for everyone;** people like Linda will never see a key. Because we shipped no keys, the demo video was recorded with pre-written AI responses; the live app uses the real APIs.

Challenges we ran into

- **Getting the law exactly right.** Sources contradicted each other: early drafts set the parent threshold at a child under 7. We encoded only rules we could cite.
- **Keeping the AI out of decisions without sounding robotic.** The AI can only fill in facts; the rules engine decides. Making that still feel human took several iterations.
- **Translating without breaking SMS.** Option numbers, codes, times and keywords must survive translation intact.
- **Making proof genuinely hard to fake.** We wrote tests that try to forge an attestation, for example turning 4 hours into 8, and made sure they fail.
- **Staying honest.** We used Iowa's real average benefit (\$168.51) rather than a rounder figure, even though it lowers our headline number.

Accomplishments that we're proud of

A complete journey you can test in two minutes: from "Hi" to a signed, verifiable attestation. A rules file a paralegal can read. Proof a caseworker can trust in one second. It works on any phone, in eight languages, with a human always one word away, and privacy is built in from the start.

What we learned

The bottleneck is **verification, not willingness**. "AI-first" doesn't have to mean "AI decides": a cited rules engine makes the tool easier to trust and to explain. And dignity has to be designed in. That means calling them "missions" and "hours that count", making Count Me In volunteers indistinguishable from everyone else, and sending nothing without a yes.

What's next for Count Me In

1. **Pilot in the DMARC network** (Nov 2026 – Mar 2027): 3–5 host sites, English and Spanish.
2. **Validate the attestation format with Iowa HHS.**
3. **Scale to Food Bank of Iowa's 55 counties**, then other states with one rules file each.

Next steps include moving the ledger to server-side PostgreSQL, loading DART's full transit feed, and funding the keys. The code is open source (MIT), with a nonprofit host and dsmHack volunteers for maintenance. We'd love Corteva to be the first corporate host site.

We don't create food. We stop it from disappearing.

Demo host sites, people and the sample letter are illustrative; no partnership is implied.